

Plant Leaf Disease Classification Using Image Processing Deep Learning-Driven Leaf Segmentation

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Abstract

The continued rise in global population is putting pressure on food production and agriculture. Additionally, the agricultural industry is facing a global labour shortage that inevitably impacts productivity and leads to increased losses in crop production. While technological innovations have helped mitigate some of these challenges, the timely and accurate identification of plant diseases remains a critical challenge. Traditional diagnostic methods, which rely on visual inspection and laboratory analysis, are often labour-intensive, time-consuming, and not easily scalable. To address this, intelligent systems and data-driven technologies have the potential to transform traditional methods of monitoring and control of plant diseases. This study presents a deep learning-based approach to detecting and identifying plant leaf diseases using integrated image processing, leaf segmentation, and data augmentation with masking-enhanced features. The goal is to create a comprehensive and more effective model that provides a scalable solution for plant disease monitoring, detection and control in the agricultural sector. The method employs HSV thresholding and EfficientNet-B1 architecture to enhance detection accuracy and reduce background bias in the analysis of a dataset of diseased plant leaves. This integrated approach methodology is designed to boost the model's focus on relevant leaf pixels, which is crucial for ensuring global food production as demand for food continues to rise.

Keywords: Plant disease classification, deep learning, image processing, leaf segmentation, data augmentation

Introduction

In November 2022, the global population reached 8 billion, and according to the United Nations, it is projected to reach 9.8 billion by 2050. This rising population has been closely linked to an increasing global demand for food (refs). To meet this demand, the global agricultural system must not only produce more food but also adapt to evolving dietary patterns and rising incomes, in line with one of the Sustainable Development Goals of the United Nations. Also, the agricultural sector is facing a global labour shortage (ref). Several factors are responsible for this, including urban migration, a growing ageing workforce, and increasing reliance on migrant and seasonal workers (ref). These limitations have resulted in reduced agricultural productivity and increased crop losses associated with low capacity utilisation and inefficiency (ref). In particular, reliance on labour has been shown to significantly limit the effectiveness of traditional plant disease detection due to human error factors (ref... González-Briones, A., Florez, S.L., Chamoso, P. et al). To address these challenges, many farmers and agricultural businesses are incorporating automation and technological advancements to mitigate the limitations of human factors or traditional labour input. Innovations such as autonomous tractors and automated irrigation systems are being adopted to reduce reliance on manual labour and enhance operational efficiency. Yet, despite these advancements, farmers face a significant challenge in identifying plant diseases in a timely and accurate manner. Early detection is crucial for preventing yield loss and ensuring the production of high-quality products. However, traditional diagnostic methods rely on visual inspection and laboratory analysis, and they require years of experience and expertise. While these methods can be effective to some extent, they are highly labour- and time-consuming and very limited in their scalability. As agricultural practice advances, intelligent systems and data-based technologies have the potential to transform disease control, driven by the growing demand for food and the need for more effective, accessible, and scalable solutions. Advances in computer vision and machine learning have enabled significant progress in the field of plant disease detection and classification. With the

application of image processing techniques on plant leaves image datasets, it has become increasingly easier to identify plant diseases early and in time to administer effective remedies and hence improve productivity.

Image Processing and Machine Learning

Image processing plays a crucial role in automating plant disease detection systems by preparing raw images for computational analysis. The primary goal of image processing is to enhance image quality in a way that facilitates accurate feature extraction and classification during the processing. This involves refining vital image details or properties, such as making leaf edges more distinct and defined, and removing background noise that could interfere with the classification process.

Machine learning, an advanced subset of artificial intelligence, builds directly on pre-processed data to classify plant diseases without explicit programming. In context, by utilising pre-processed images as training data, machine learning models can adapt to complex visual patterns, thereby automating, optimising, and understanding the image to interpret and classify plant diseases without explicit programming. This enables the classification of unseen sample sets for plant diseases. This offers farmers efficient and accurate diagnostic solutions. This approach enhances decision-making while also playing a vital role in promoting food security, reducing dependency on expert labour, increasing resource efficiency, and facilitating early detection of climate-driven diseases.

Within this field, deep learning is a subset of machine learning that uses neural networks with multiple layers to learn complex patterns. However, it is computationally expensive, requires large datasets, and high-performance computing resources. An alternative approach, statistical machine learning and classical image processing, is effective with smaller datasets and is less computationally expensive. This approach focuses on statistical techniques and algorithms such as linear regression and decision trees, which facilitate scalability depending on farm size, available resources, and project scope.

This study builds on these approaches by developing and evaluating a deep learning-driven approach to plant leaf disease classification that integrates image processing, leaf segmentation, and data augmentation with masking-enhanced features. It utilises a plant disease dataset and combines HSV thresholding with an EfficientNet-B1 architecture to train and evaluate classification models. This integrated approach aims to enhance disease detection accuracy, mitigate background bias, and scale effectively. It holds significant potential in safeguarding global food production as global population and food demand continue to increase.

Methodology

This section outlines the approach used to develop a deep learning model for plant disease classification based on image data. The methodology includes dataset collection, pre-processing, model architecture selection and training strategy.

1.1 Datasets

There are various datasets available; the most common and recently used one is PlantVillage. The PlantVillage dataset, a publicly available dataset for plant leaf disease detection, has been utilised and verified by various researchers.

It consists of 54,303 images, featuring 14 plant species in total and 38 categories of different species, including both healthy and diseased states. The dataset also includes a class for background images without leaves. The background images

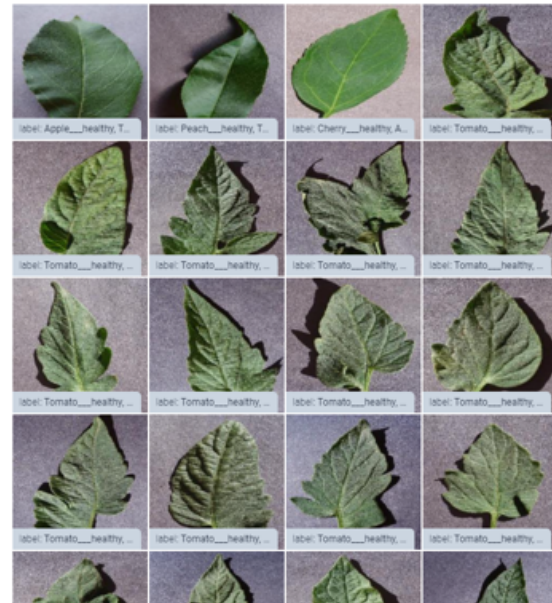


Figure 1: PlantVillage Dataset Images

serve as a negative/control class, reducing false positives and helping the model distinguish between leaf and non-leaf content. Furthermore, the inclusion enhances the model's robustness in handling real-world images that may include various types of obstructions or partial leaves, as opposed to full leaves.

1.2 Data Splitting

To prepare the dataset for the model, the dataset was divided into three mutually exclusive subsets: training (75%), validation (15%), and test (10%) subsets.

- **Training set:** Used to teach the model by learning from labelled plant disease images.

During this phase, the model learns patterns by adjusting its weights to minimise prediction

errors. In a neural network, weights are numerical values associated with the connections between two neurons (or nodes). These weights determine the important data in the input image while disregarding irrelevant data. As an input is passed through a connection, it is multiplied by the corresponding weight and passed to the next neuron, determining the influence one neuron has over the next.

To improve generalisation, the model is initialised with pretrained ImageNet weights, which helps to speed up the training process. These pretrained weights provide a foundational source of generalised visual features from prior training on a large-scale dataset, rather than requiring the model to learn from scratch. This approach improves resource efficiency while enhancing accuracy and performance. Then, the final classification layer, added to adapt the model for the 39 plant disease categories, is randomly initialised using He/Kaiming initialisation to prevent issues such as vanishing or exploding gradients.

Throughout training, the model refines these weights through backpropagation and gradient descent. This process involves comparing predicted output to the expected output (in this case, the correct plant diagnosis), calculating the error, and adjusting the weights to reduce error over time and emphasise the more important connections.

- **Validation set:** Used to evaluate the model at the end of each training epoch, without influencing weight updates. This enables the monitoring of the model's performance on unseen data during training, aiding in hyperparameter tuning and selecting the optimal model state based on validation accuracy. The validation process also helps prevent overfitting, which occurs when the model is fitted too closely to the training data, thereby introducing noise that limits the model's ability to generalise to unseen data, resulting in a gap between the training and validation accuracy. Conversely, the validation results can

reveal underfitting, which occurs when the model is too simple and fails to learn the data structure and capture the underlying patterns.

- **Testing set:** An unseen set used after the training and validation phases are complete. It acts as a final unbiased assessment of the model's performance on completely unseen data, reflecting its expected real-world accuracy.

To ensure reproducibility and consistent splits, a fixed random seed was applied when dividing the data, ensuring that the dataset split occurs the same way every time the code is run. After splitting, all subsets were loaded using PyTorch's 'DataLoader'. For the training loader, data shuffling was enabled to reduce sampling bias between epochs, while the validation and test loaders preserved a fixed order.

1.3 Image Preprocessing and Augmentation

A series of data transformations was applied to each subset to enhance performance, normalise and augment the dataset for generalisation and robustness.

For the training subset, a set of augmentations was implemented to mitigate the risk of overfitting and increase the diversity of the data.

- **RandomResizedCrop:** Instead of uniform resizing, a random portion of the image was cropped and resized to 240x240 pixels. This helps train the models' ability to identify key features of plant diseases, regardless of whether they are in the centre of the image or the background.
- **RandomHorizontalFlip and RandomRotation(15):** This randomly flips the image horizontally with 50% probability and randomly rotates the image by an angle between -15 ° and +15°. These augmentations help the model recognise a disease on a leaf, whether it appears on

the left or right side of the leaf. This allows it to become invariant to the left-right orientation of the plant leaf image and prevents overfitting on fixed orientations.

- **Normalise (imagenet_mean, imagenet_std):** The image pixel values were normalised by subtracting the mean ([0.485,0.456,0.406]) by the standard deviation ([0.229,0.224,0.225])

For the validation and test subsets, the argumentations applied were deterministic; the same procedure was applied to every image to ensure the model's consistency and fairness. The arguments were like training sets' transformations, but instead of 'random' transformations, resizing to 276×276 and centre cropping to 240×240 pixels were applied, followed by the same normalisation.

1.4 Leaf Region Segmentation

To reduce background bias, each image is pre-processed by identifying the leaf region using colour-based segmentation in the HSV colour space. To do this, the following steps were taken for mask Generation:

1. Convert the RGB image to the HSV colour space

Given an input image, load the image using PIL (Python Imaging Library), then convert to an OpenCV BGR NumPy array for masking:

$$I_{RGB}(x,y) = [R(x,y), G(x,y), B(x,y)]$$

$$I_{BGR}(x,y) = BGR(I_{RGB}(x,y))$$

Then convert it to the HSV colour space:

$$I_{HSV} = HSV(I_{BGR}(x,y))$$

Where each pixel is represented as

$$I_{HSV}(x,y) = [H(x,y), S(x,y), V(x,y)]$$

2. Leaf Region Masking:

The empirical lower bound (T_l) and upper bound (T_u) are defined in HSV space:

$$T_l = [25,40,40]$$

$$T_u = [85,255,255]$$

They were specially set to preserve symptomatic leaf areas that may be yellow or brown, such as Cercosporin leaf spots, to improve precision and accuracy, going further than simply looking for only green hues.

The leaf mask is then obtained using a thresholding technique.

$$M(x,y) = \begin{cases} 1, & T_l \leq I_{HSV}(x,y) \leq T_u \\ 0, & otherwise \end{cases}$$

$$I_{leaf}(x,y) = I_{BGR}(x,y) \cdot M(x,y)$$

And then an inverse mask for non-leaf regions:

$$I_{nonleaf}(x,y) = I_{BGR}(x,y) \cdot (1 - M(x,y))$$

3. Context-Preserving Blending

To keep the pixel with colours ($M=1$), blend 90% leaf region and 10% non-leaf with a blending coefficient of $\alpha = 0.9$

$$I_{blended}(x,y) = \alpha \cdot I_{leaf}(x,y) + \beta \cdot I_{nonleaf}(x,y)$$

4. Covert Back to RGB and Augment

Convert $I_{blended}$ from BGR back to RGB, then to PIL for PyTorch.

$$I_{final} = RGB(PIL(I_{blended}(x,y)))$$

Then apply the augmentations and normalisation depending on the subset.

$$I_{aug} = Transform(I_{final})$$

Full composite Formula:

Given input image (I_1) and binary mask (M):

$$I_{preprocessed} = Transform(\alpha \cdot I_1 \odot M + (1 - \alpha) \cdot I_1 \odot (1 - M))$$

Overall, leaf segmentation removes background features, reduces bias, and improves the model's focus on relevant leaf pixels, resulting in overall better performance.

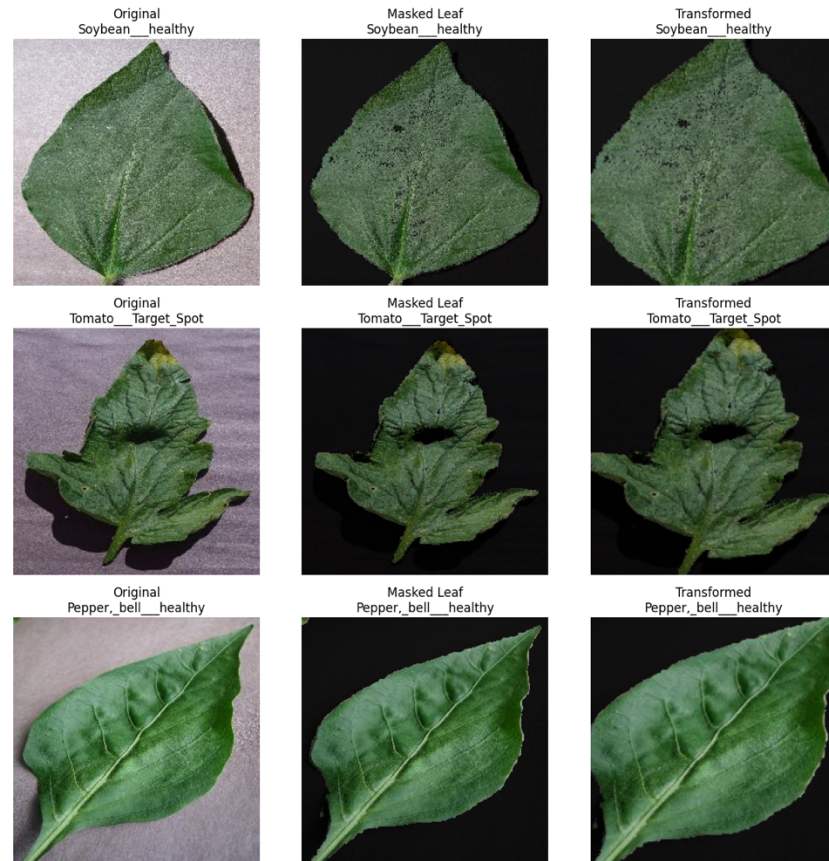


Figure 2: Effect of Leaf Segmentation and Augmentation

1.5 Model Architecture and Training Procedure

The model architecture used was EfficientNet-B1, a pretrained Convolutional Neural Network (CNN) model trained on the ImageNet 1K dataset (1.2 million images, 1,000 classes). EfficientNet-B1 offers better efficiency and accuracy compared to traditional CNNs, such as ResNet or VGG, while balancing model complexity with computational efficiency.

The following parameters were used during training:

- Optimiser: Stochastic Gradient Descent (SGD) with learning rate of $1e-3$, momentum of 0.9 and weight decay of $1e-4$ for regularisation
- Loss Function: Cross-Entropy Loss is well-suited for multi-class classification, such as plant disease classification.
- Batch Size: 64 images per iteration was used on all loaders, with four workers, and pin memory enabled for improved GPU throughput
- Gradient Scaling: Automatic Mixed Precision for faster training and efficient GPU usage.

The model was trained for 10 epochs and set to `train()` mode for backpropagation. After each epoch, the model was evaluated on the validation set using `eval()` mode. During this process, the loss and accuracy are tracked and stored for both phases across batches for further analysis.

Results and Discussion

The EfficientNet-B1 model achieved a final test accuracy of 98.45% and a test loss of 0.0538, indicating its ability to generalise to unseen data and classify plant diseases.

Figure 3 shows a positive and consistent trend in the training and validation accuracy over 10 epochs. In the first epoch, the model achieved **54.18%** training accuracy and **81.17%** validation accuracy, as seen in Figure 4. This suggests that the pre-processing pipeline and leaf segmentation masking were highly effective in extracting relevant disease features from the training subset dataset. Notably, as the training and validation accuracy increased, the training and validation loss decreased. The narrowing gap between the training and validation lines indicates a balance in memorisation, learning, and generalisation without significant overfitting. This underscores the model's ability to learn the plant features that distinguish plant-specific diseases, effectively aided by strong data augmentation and hyperparameters.

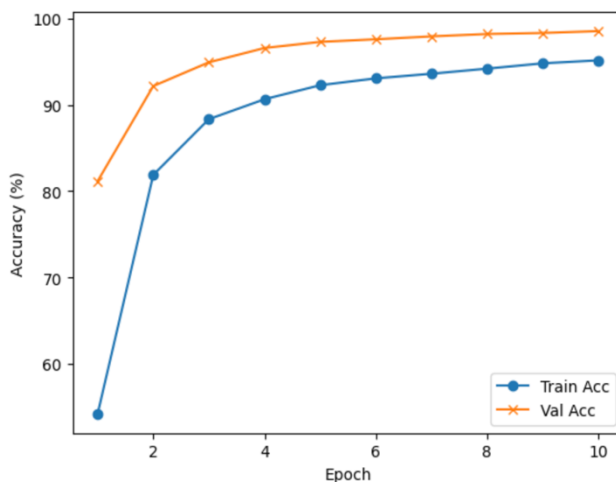


Figure 4: Training vs. Validation Accuracy (10 Epochs)

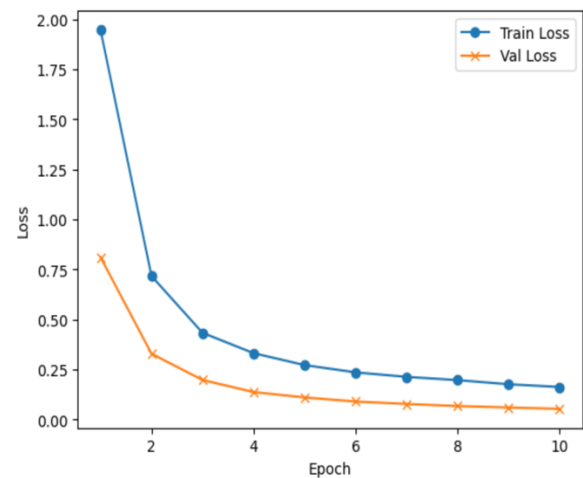


Figure 3: Training vs. Validation Loss (10 Epochs)

Model Performance Analysis

The model achieved a macro and weighted F1-score of 0.98 across all 38 classes. The stronger-performing classes included Apple Cedar Apple Rust, Grape Leaf Blight, Orange Huanglongbing, Healthy Grapes, and Squash Powdery Mildew, each with $F1 = 1.00$, no false positives (FP), and no false negatives (FN). This could be due to having distinctive visual symptoms and a high amount of training data, making classification and learning easier.



Figure 5: Corn Cercospora leaf spot

On the other hand, the weaker-performing classes included Corn Cercospora Leaf Spot ($F1 = 0.90$), Potato Healthy ($F1 = 0.89$), and Corn Northern Leaf Blight ($F1 = 0.92$). This could occur due to confusion with visually similar disease patterns, for example, as shown in Figures 5 and 6. Corn Cercospora Leaf Spot and Corn Northern Leaf Blight both have elliptical lesions that are long and narrow. Another factor could be mild discolouration or shadows in the image that



Figure 6: Corn Northern Leaf Blight

could be misclassified as a disease. Another factor is limited and imbalanced training data for certain classes. For example, Potato Healthy has only 152 samples, whereas Potato Early Blight has 1,000 samples, thereby reducing the model's ability to discern patterns.

From a practical perspective, the results indicate that a CNN can be trained to classify plant diseases based on leaf images with high reliability, provided it is trained properly, making it suitable for integration into real-world agricultural sectors. However, even with 98% accuracy, it is important to note

that the dataset used was sourced from PlantVillage, where images have similar lighting and a controlled background, which does not represent the more variable field conditions.

Improving real-world applicability could include expanding the HSV threshold ranges and incorporating colour jitter augmentation to accommodate varying field lighting conditions and various symptom variations. Testing with more on-site leaf images could also help improve model training. Additionally, testing more advanced segmentation architectures, such as U-Net, could reduce reliance on colour cues while providing more accurate and robust segmentation, encouraging future learning and making the model more adaptable to real-world plant images.

Limitations and Challenges

In developing the plant disease classification model, an accuracy of 98% was achieved; however, several limitations and challenges were identified that impact the model's reliability and real-world applicability. Recognising these limitations is essential, as it highlights areas where the model might not perform well outside of controlled and familiar environments and offers development opportunities. Addressing these limitations can enhance the accuracy, robustness, and applicability of the model in real-world agricultural settings.

Dataset Constraint

The key challenge is dataset constraints. To truly test the effectiveness when deployed in real-world scenarios, it is vital to train the model on a larger and more diverse dataset that has a better representation of real-world field conditions. This has a direct impact not only on the model's performance but also on its robustness and capability to generalise a wide range of species. Otherwise, the model is at risk of overfitting, underfitting and poor generalisation to unseen data. An approach to solving this problem is through data sharing initiatives.

Class Imbalance

This limitation is an issue for certain classes that are underrepresented compared to others. It directly affects the model's learning process, performance, and fairness due to the imbalance. It could lead to biased predictions towards the majority classes, resulting in misleading evaluation metrics and higher false negative rates. This highlights the importance of the F1 score, as it equally weights all classes. To address these issues, exploring resampling methods could ensure that the model covers all classes more evenly.

Segmentation Limits

This approach may reduce background and enhance leaf classification, but its effectiveness is contingent upon specific lighting conditions. In varying lighting conditions, it may result in under-segmentation (loss of leaf areas) or over-segmentation (inclusion of background). Furthermore, in real-world field images, soils like leaf tissue could lead to false positives during segmentation. Additionally, the approach lacks adaptability to seasonal or regional variations of leaf pigmentation, which could be attributed to climate or nutritional deficiencies.

Uniform Backgrounds

The PlantVillage dataset was captured in a controlled environment with a uniform background that was often grey. While the model performed well with this dataset, its accuracy is likely to decrease when working with images featuring complex backgrounds in real-world agricultural settings. This makes the model less robust when exposed in real time, due to the high variability in actual farming environments. This could lead to misclassification of leaves or the model picking up back cues rather than disease features. To address this, incorporating field-acquired images in the training set is vital to ensure background diversity, and training segmentation models that can differentiate leaves from backgrounds without colour thresholds would be more effective.

Resource Constraint

Although the model has high accuracy, its application in agricultural settings may be inhibited by resource limitations, particularly in low to middle-income regions and on small to medium-sized farms. Deep learning requires large datasets for optimal efficiency, which in turn requires large computational power. This could result in farmers encountering hardware limitations, such as limited access to high-performance computing devices, insufficient RAM, or GPUs that are not capable of running complex

CNN models in real time, which can hinder deployment. Additionally, image quality is key in capturing disease patterns and details, all of which are essential for accurate plant disease detection. Low image quality can obscure vital features that may affect leaf segmentation, processing, and classification.

Model Interpretability

While the model, if well-trained, can present accuracy in disease prediction, farmers without prior experience with AI-based tools may require training not only to utilise the model itself but also to interpret the data. For farmers to fully benefit from the system, they need to understand the reason behind the model's prediction, especially since agricultural environments are highly variable. Through understanding, farmers can trust the model's output and adapt the system to local conditions. Furthermore, this knowledge could help them to diagnose model errors and potentially aid in the development of preventive and management practices.

Future Directions

To further enhance the effectiveness of the plant disease classification model, it is essential to explore future directions that go beyond detection alone. These future directions aim not only to improve model robustness and reliability but also to deliver meaningful economic and agricultural impacts in the long term as follows.

Economic benefit

Automated plant disease detection offers economic advantage to farmers and the agricultural industry, particularly considering the global labour shortage in the farming sector. With fewer experienced workers available for manual crop monitoring and traditional detection methods, automated systems can fill this gap with minimal human intervention. Early detection of plant diseases can mitigate crop losses, facilitate timely prevention, and prevent the spread of infections before they become more severe. Early intervention not only prevents economic losses but also promotes enhanced resource efficiency. Farming can reduce input costs by concentrating on targeted treatments while limiting excessive use of pesticides and fertilisers. Collectively, these factors have the potential to revolutionise traditional disease management through data-driven decision-making and forecasting.

Advancement in Crop Recommendation

Crop recommendation involves the application of machine learning to provide advanced decision support to farmers with respect to suitable crop selection and farming practices under specific local conditions or variables. It could be applied to plant disease detection through the integration of predictive analytics and environmental monitoring. Together, it could go further, not only detecting plant issues but also recommending optional crops based on disease outbreak data. For example, if a

late blight is constantly being detected in a region, the system could recommend an alternative crop or treatment.

Conclusion

In conclusion, the application of deep learning and image processing for plant leaf disease classification provides an effective way to improve disease control and enhance food security, while reducing reliance on expert labour. Through this study, a deep learning model that successfully integrates leaf segmentation and data augmentation to strengthen disease detection accuracy and mitigate background bias has been developed. Utilising an EfficientNet-B1 architecture, the model achieved a high test accuracy of 98.45%, demonstrating its strong ability to generalise to unseen data.

While the model showed promising results, the research also highlighted several limitations, including constraints from a limited and controlled dataset, as well as the potential need for greater model interpretability to ensure real-world applicability. Continued research and exploration into these challenges can further enhance the potential of this technology to transform the future of agriculture. Nonetheless, the study lays the groundwork for the advancement of smart and resilient agricultural practices that will have a significant positive and lasting impact on improved crop production for global food supply and security.

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Tables

Table 1: Image Counts by Plant Type and Associated Diseases

Plant	Disease Name	No.Images
Apple	<i>Healthy</i>	1645
	<i>Diseased: Scap</i>	630
	<i>Diseased: Black Rot</i>	621
	<i>Diseased: Cedar Apple Rust</i>	275
Blueberry	<i>Healthy</i>	1502
Cherry	<i>Healthy</i>	854
	<i>Diseased: Powdery Mildew</i>	1052
Corn	<i>Healthy</i>	1162
	<i>Diseased: Cercospora leaf spot</i>	513
	<i>Diseased: Common rust</i>	1192
	<i>Diseased: Northern Leaf Blight</i>	985
Grapes	<i>Healthy</i>	423
	<i>Diseased: Black rot</i>	1180
	<i>Diseased: Esca (Black Measles)</i>	1383
	<i>Diseased: Leaf blight (Isariopsis)</i>	1076
Orange	<i>Diseased: Haunglongbing (Citrus Greening)</i>	5507
Peach	<i>Healthy</i>	360
	<i>Diseased: Bacterial Spot</i>	2297
Pepper Bell	<i>Healthy</i>	1478
	<i>Diseased: Bacterial Spot</i>	997
Potato	<i>Healthy</i>	152
	<i>Diseased: Early blight</i>	1000
	<i>Diseased: Late blight</i>	1000
Raspberry	<i>Healthy</i>	371
Soybean	<i>Healthy</i>	5096
Strawberry	<i>Healthy</i>	456
	<i>Diseased: Leaf Scorch</i>	1109
Tomato	<i>Healthy</i>	1591
	<i>Diseased: Bacterial spot</i>	2127
	<i>Diseased: Early blight</i>	1000
	<i>Diseased: Late blight</i>	1909

	<i>Diseased: Leaf Mold</i>	952
	<i>Diseased: Septoria leaf spot</i>	1771
	<i>Diseased: Two-spotted spider mite</i>	1676
	<i>Diseased: Target Spot</i>	1404
	<i>Diseased: Yellow Leaf Curl Virus</i>	5357
	<i>Diseased: Tomato mosaic virus</i>	373
Background without leaves		1143

Table 2: Classification Report (Precision, Recall, F1-Score, Support)

Class	Precision	Recall	F1-Score	Support
<i>Apple___Apple_scab</i>	1.0	0.97	0.98	66
<i>Apple___Black_rot</i>	0.97	1.0	0.99	66
<i>Apple___Cedar_apple_rust</i>	1.0	1.0	1.0	32
<i>Apple___healthy</i>	0.99	0.98	0.99	176
<i>Background_without_leaves</i>	1.0	0.98	0.99	114
<i>Blueberry___healthy</i>	0.99	1.0	0.99	166
<i>Cherry___Powdery_mildew</i>	0.99	0.99	0.99	120
<i>Cherry___healthy</i>	1.0	0.99	0.99	79
<i>Corn___Cercospora_leaf_spot Gray_leaf_spot</i>	0.87	0.92	0.9	66
<i>Corn___Common_rust</i>	0.98	1.0	0.99	108
<i>Corn___Northern_Leaf_Blight</i>	0.95	0.9	0.92	107
<i>Corn___healthy</i>	0.99	1.0	1.0	120
<i>Grape___Black_rot</i>	0.97	0.98	0.98	111
<i>Grape___Esca_(Black_Measles)</i>	0.99	0.98	0.98	150
<i>Grape___Leaf_blight_(Isariopsis_Leaf_Spot)</i>	1.0	1.0	1.0	115
<i>Grape___healthy</i>	1.0	1.0	1.0	47
<i>Orange___Haunglongbing_(Citrus_greening)</i>	1.0	1.0	1.0	522
<i>Peach___Bacterial_spot</i>	0.99	0.99	0.99	225
<i>Peach___healthy</i>	0.97	0.97	0.97	36
<i>Pepper,_bell___Bacterial_spot</i>	0.99	0.99	0.99	96
<i>Pepper,_bell___healthy</i>	0.99	0.99	0.99	156
<i>Potato___Early_blight</i>	0.98	0.97	0.97	91
<i>Potato___Late_blight</i>	0.95	0.94	0.95	88
<i>Potato___healthy</i>	0.92	0.86	0.89	14
<i>Raspberry___healthy</i>	1.0	1.0	1.0	28

<i>Soybean__healthy</i>	<i>0.99</i>	<i>1.0</i>	<i>0.99</i>	<i>515</i>
<i>Squash__Powdery_mildew</i>	<i>1.0</i>	<i>1.0</i>	<i>1.0</i>	<i>189</i>
<i>Strawberry__Leaf_scorch</i>	<i>0.99</i>	<i>1.0</i>	<i>1.0</i>	<i>119</i>
<i>Strawberry__healthy</i>	<i>1.0</i>	<i>1.0</i>	<i>1.0</i>	<i>49</i>
<i>Tomato__Bacterial_spot</i>	<i>0.97</i>	<i>0.98</i>	<i>0.98</i>	<i>223</i>
<i>Tomato__Early_blight</i>	<i>0.92</i>	<i>0.9</i>	<i>0.91</i>	<i>93</i>
<i>Tomato__Late_blight</i>	<i>0.96</i>	<i>0.98</i>	<i>0.97</i>	<i>206</i>
<i>Tomato__Leaf_Mold</i>	<i>1.0</i>	<i>0.97</i>	<i>0.98</i>	<i>88</i>
<i>Tomato__Septoria_leaf_spot</i>	<i>0.99</i>	<i>0.97</i>	<i>0.98</i>	<i>176</i>
<i>Tomato__Spider_mites Two-spotted_spider_mite</i>	<i>0.99</i>	<i>0.98</i>	<i>0.98</i>	<i>167</i>
<i>Tomato__Target_Spot</i>	<i>0.94</i>	<i>0.94</i>	<i>0.94</i>	<i>140</i>
<i>Tomato__Tomato_Yellow_Leaf_Curl_Virus</i>	<i>0.99</i>	<i>0.99</i>	<i>0.99</i>	<i>496</i>
<i>Tomato__Tomato_mosaic_virus</i>	<i>0.95</i>	<i>1.0</i>	<i>0.98</i>	<i>41</i>
<i>Tomato__healthy</i>	<i>0.99</i>	<i>0.99</i>	<i>0.99</i>	<i>144</i>
<i>Accuracy</i>		<i>0.98</i>	<i>0.98</i>	<i>5545</i>
<i>Macro Avg</i>	<i>0.98</i>	<i>0.98</i>	<i>0.98</i>	<i>5545</i>
<i>Weighted Avg</i>	<i>0.98</i>	<i>0.98</i>	<i>0.98</i>	<i>5545</i>